

Lead-Lag Networks in Metal Trading

Presentation Materials — proxy network graphs, sensitivity analysis & practical implications

Period: **2020-01-01** → **2025-01-01** · Frequency: **daily** · Assets: **8** · Observations: **1 305**
Companion files: [notebook.ipynb](#) · [REPORT.md](#) · figures in [outputs/presentation/](#)

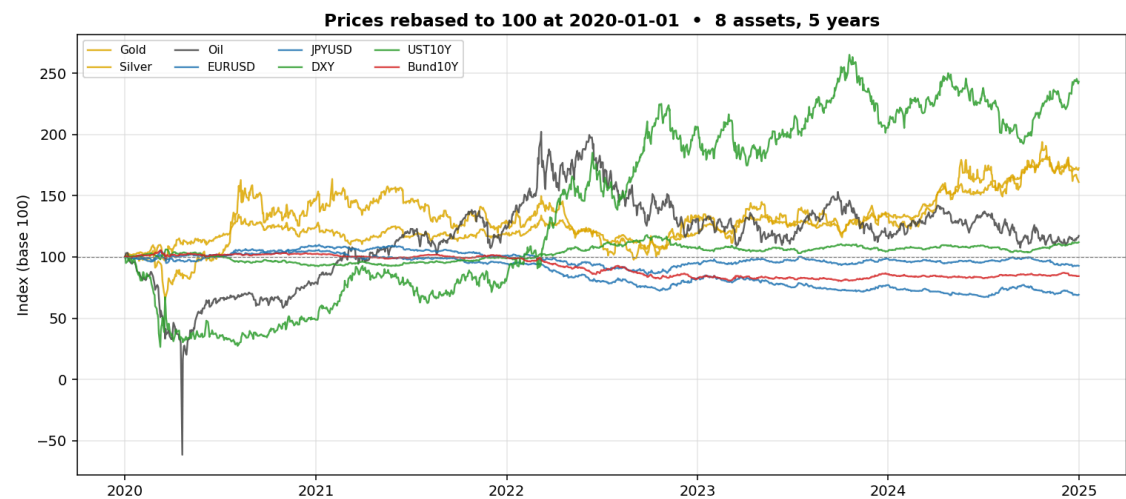
1 · Executive summary

We screen eight cross-asset price series (precious metals, energy, FX, US/EU rates) for **directional lead-lag relationships** using **Dynamic Time Warping** and an **information-theoretic transfer graph**. Three actionable conclusions:

#	Finding	Trading / risk use
1	Gold leads Bund10Y (+7 d), JPY (+6 d), DXY (+5 d)	Directional filter on safe-haven FX / rates
2	UST10Y → Bund10Y transmission ≈ 11 days	Spread / curve trades on the Fed→ECB cycle
3	JPY ↔ Bund is the tightest pair (DTW = 9.3)	Structural carry-trade node — early warning of regime shift if it breaks

The full universe organises into **three natural clusters** (Metals · USD/Rates · EUR/JPY/Bund), visible both on the DTW distance matrix and on the proxy network graph below.

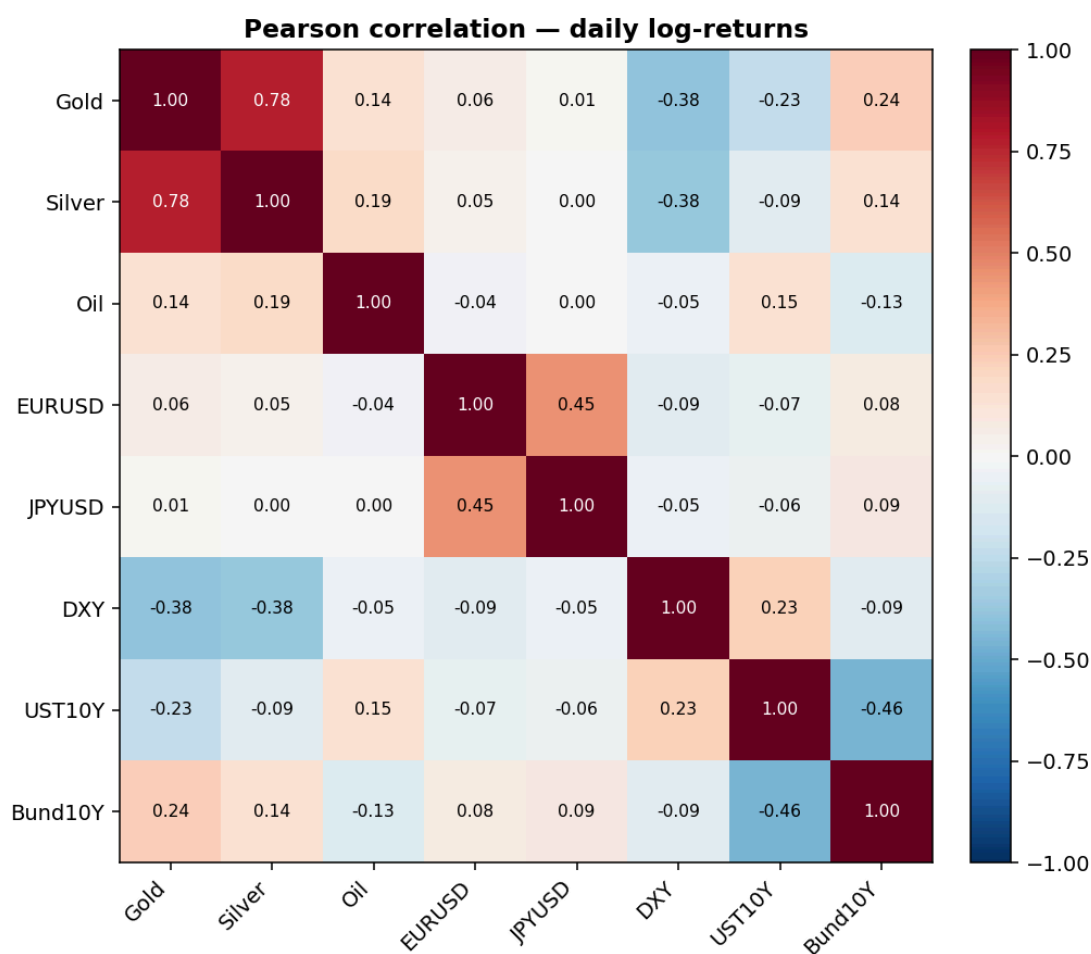
2 · The universe at a glance



Eight assets rebased to 100 on 2020-01-01. Gold and the EUR-leg rates rallied (Gold +70 %), DXY stayed flat, JPYUSD and EURUSD lost ground. The negative WTI spike in April 2020 (CL=F front-month settled below 0) is preserved — it is a real market event, not a data artefact.

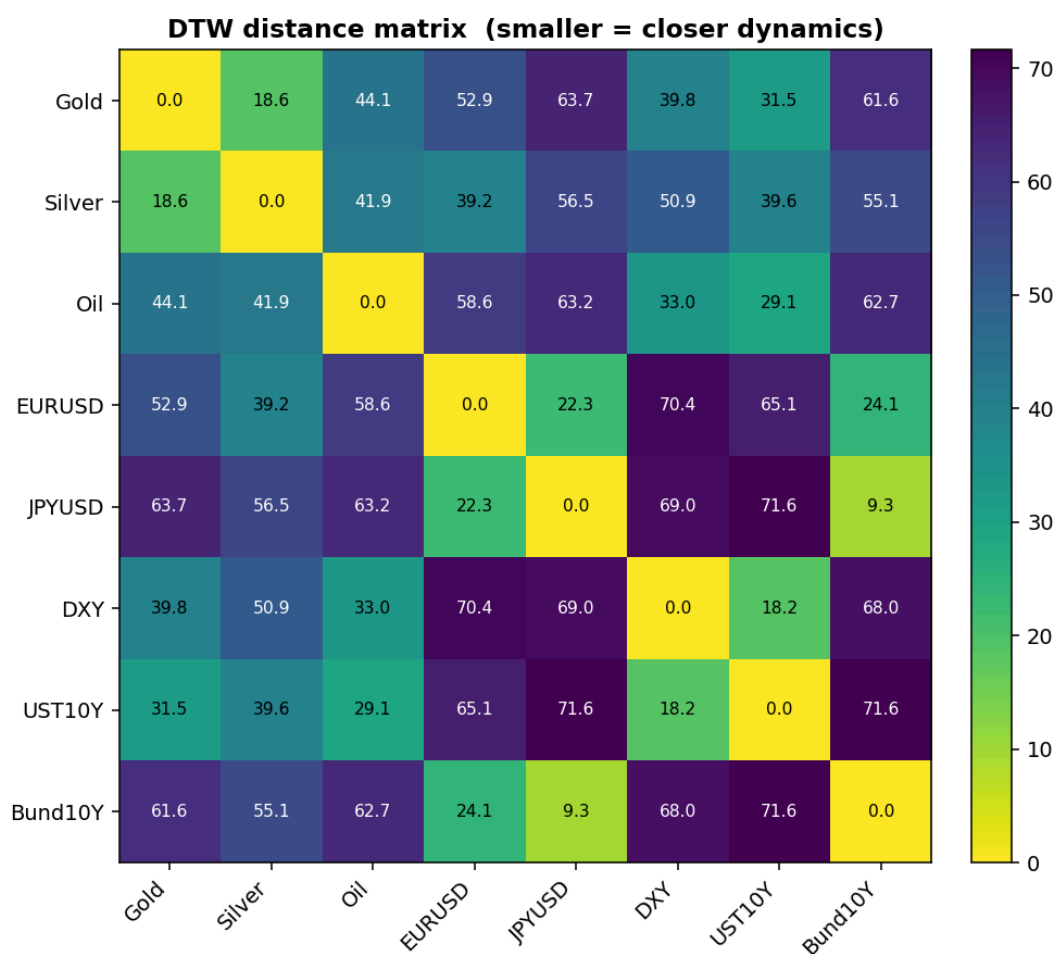
Cluster	Tickers	Story 2020-2025
Metals	Gold (GC=F), Silver (SI=F)	Risk-off + USD weakening → rally
Energy	Oil (CL=F)	COVID negative print → recovery
FX	EURUSD, JPYUSD	USD strength, BoJ divergence
USD / Rates	DXY, UST10Y	Fed tightening cycle
EUR / Rates	Bund10Y (IS0L.DE)	ECB lag vs. Fed

3 · Linear correlations — what we *don't* see



Pearson correlations (log-returns) confirm the obvious blocks (**Gold-Silver 0.78**, metals ↔ **DXY -0.38**, **UST10Y ↔ Bund10Y -0.46**, EUR ↔ JPY 0.45) but they are **symmetric and contemporaneous**. They cannot tell us *who moves first*. That motivates the DTW + lead-lag pipeline.

4 · DTW distances — three clusters emerge



Reading the matrix (smaller = more similar dynamics):

Pair	DTW distance	Interpretation
JPYUSD ↔ Bund10Y	9.3 ★	Tightest pair → JPY funding into EUR govt bonds
DXY ↔ UST10Y	18.2	The USD/rates dipole
Gold ↔ Silver	18.6	Precious-metals block
EURUSD ↔ JPYUSD	22.3	Shared "USD strength" factor
EURUSD ↔ Bund10Y	24.1	EUR macro complex

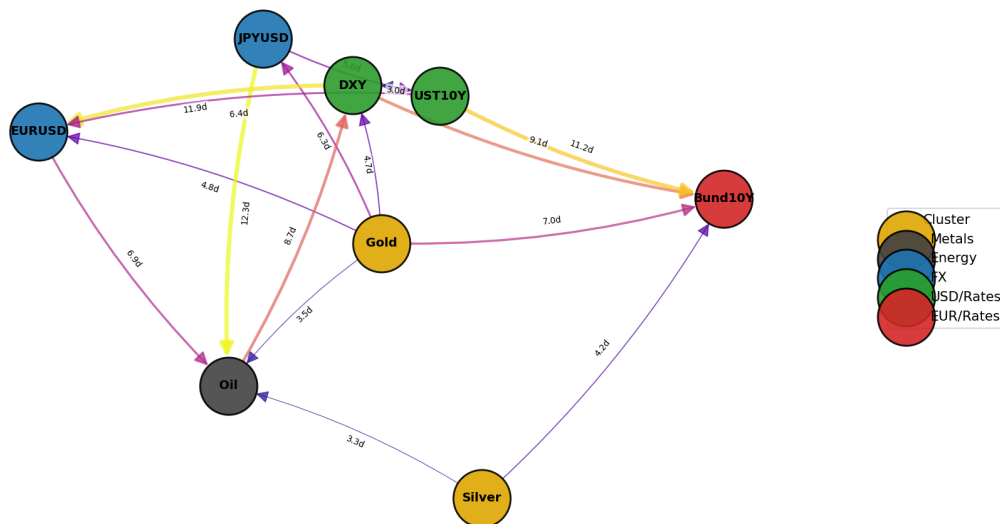
The clusters in the report's clustermap ([03_dtw_clustermap.png](#)) line up with these pairs.

5 • Proxy lead-lag network ★

The **directed graph** below is the centrepiece of this presentation. Each edge $A \rightarrow B$ means asset A's price moves precede asset B's by **at least 3 trading days** (mean signed deviation of the DTW

warping path from the diagonal). Edge labels carry the inferred lag in days; line width / colour encode lag magnitude.

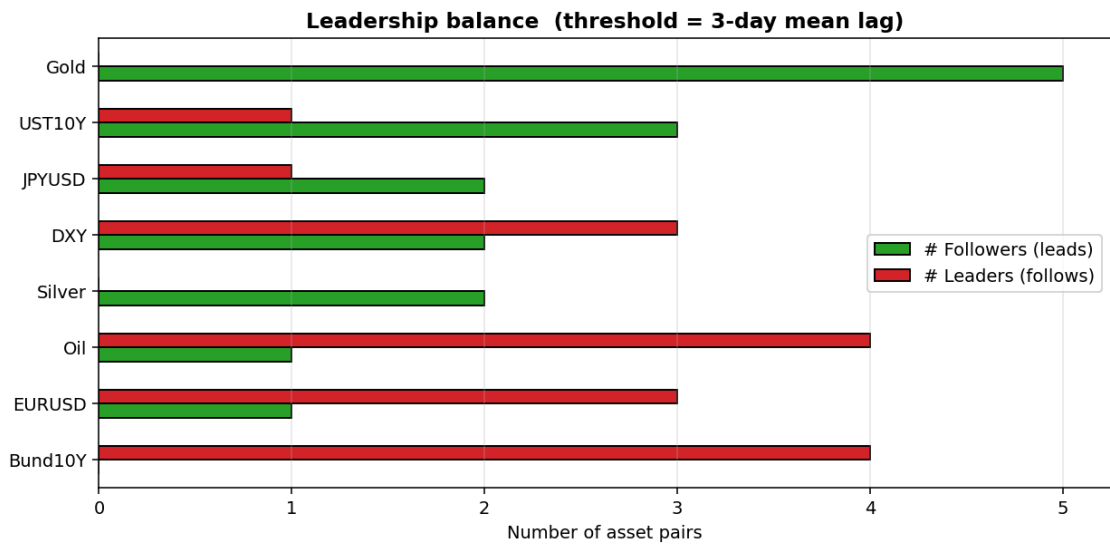
Proxy lead-lag network • edge A→B means A leads B by ≥ 3 days (label = mean lag from DTW path)



Structural reading

- **Gold is the system's most prolific *out-going* node.** Arrows leave Gold toward every cluster — Bund10Y (+7.0 d), JPYUSD (+6.3 d), DXY (+4.7 d), EURUSD (+4.8 d), Oil (+3.5 d).
- **UST10Y → Bund10Y (+11.2 d)** is the longest lag in the system and the clearest macro story: Fed re-pricing flows into the European curve with roughly two trading weeks of delay.
- **DXY → EURUSD (+11.9 d)** is mechanical (EUR has 57 % weight in DXY) but the persistence is informative: the dollar index moves *before* the cleanest bilateral pair stabilises.
- **JPY → Oil (+12.3 d)** is the longest non-mechanical edge — JPY weakness anticipates oil rallies, consistent with JPY's risk-asset funding role.
- **No edge enters Gold above the 3-day threshold** → in this sample, Gold is *only a sender*.

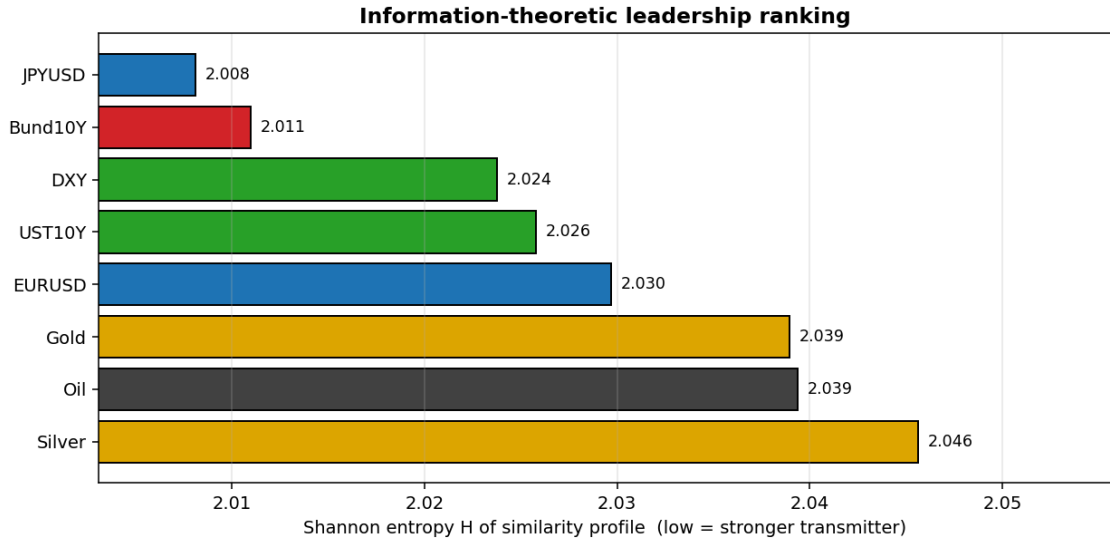
Aggregate leadership balance



For each node, we count outgoing edges (it leads N other assets) and incoming edges (it follows N others). **Gold is the only pure source** (5 / 0). **Bund10Y is a pure sink** (0 / 4). Oil and EURUSD are mostly receivers — they integrate signals from the rest of the system.

6 • Information-theoretic ranking (entropy transfer graph)

The DTW distance matrix is converted into a row-stochastic similarity profile per asset, then summarised by Shannon entropy H (low H = sharply peaked profile = strong transmitter).



Rank	Asset	H	Role
1 ★	JPYUSD	2.008	Strongest transmitter — hub of the EUR/JPY/Bund cluster
2	Bund10Y	2.011	Co-hub with JPY

3	DXY	2.024	USD macro switchboard
8	Silver	2.046	Most reactive / diffuse — pure receiver

The entropies are tight (2.01 – 2.05), so all assets share substantial mutual information. The ordering still discriminates two practical groups: **transmitters (JPY, Bund, DXY)** and **receivers (Silver, Oil, Gold)**.

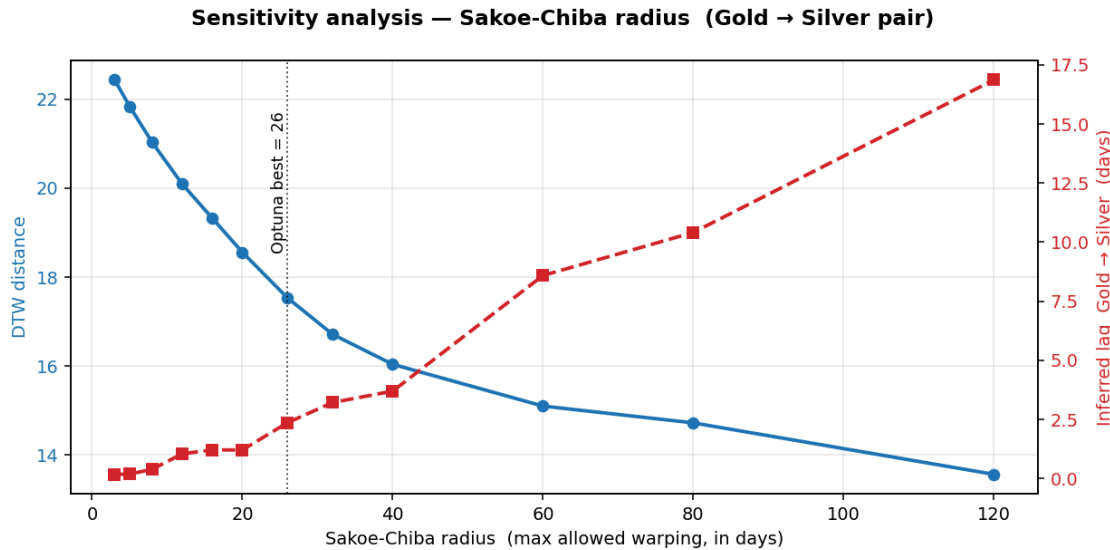
Why does Gold rank as a receiver here but appears as a leader in the DTW graph? The DTW lead-lag is **temporal**: who moves first. The entropy metric is **structural**: how concentrated the similarity profile is. Gold has a broad similarity profile (touches metals, FX and rates) — so its entropy is high. But its movements still arrive first in time. The two views are complementary, not redundant.

The full transfer-entropy graph layout (MDS embedding) is in [outputs/06_entropy_graph.png](#).

7 · Sensitivity analysis — model parameters

7.1 Sakoe-Chiba radius (DTW window)

The Sakoe-Chiba band limits how far the warping path can deviate from the diagonal — it is the **single most impactful hyperparameter** of the DTW lead-lag pipeline. We sweep it on the Gold→Silver pair (Optuna's tuning target) and track both the DTW distance and the inferred mean lag.

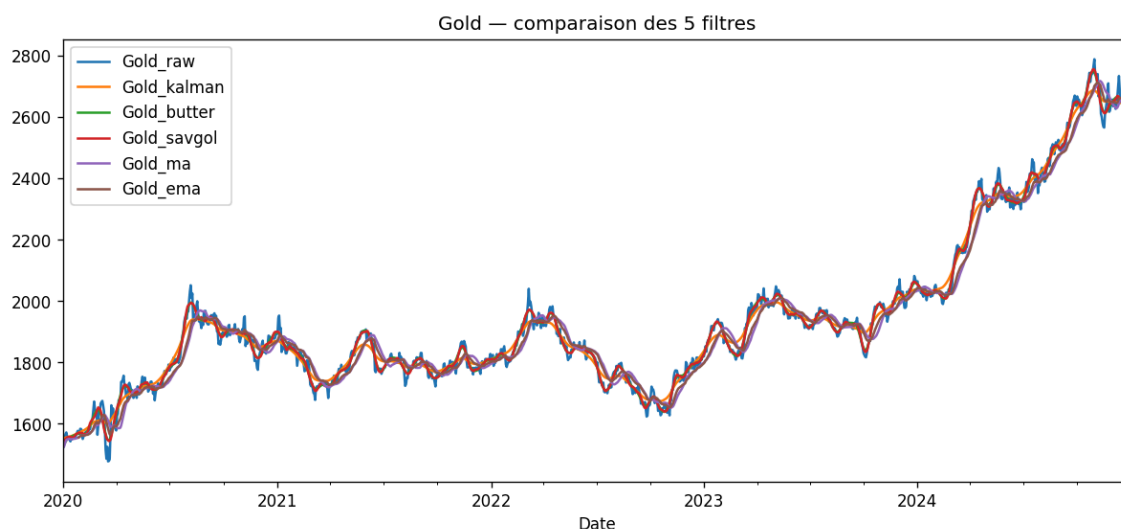


Radius	DTW distance	Inferred lag (d)	Comment
3	22.4	≈ 0	Path forced to diagonal → DTW collapses to Euclidean, lag erased
26 ★	17.5	≈ 2.5	Optuna optimum on RMSE CV: best trade-off between flexibility and over-warping

60	15.1	8.5	Path drifts → spurious lag inflation
120	13.5	17	Unbounded warping → meaningless lag

Take-away. A small radius (< 10 d) under-fits and hides the lead-lag. A large radius (> 60 d) over-fits and inflates the lag artificially. The Optuna optimum ($r = 26$) sits exactly at the elbow of the distance curve — the lag is stable in a $[20, 40]$ neighbourhood, which gives the result robustness.

7.2 Filter choice (Task 2)



Five filters were benchmarked on Gold (Kalman, Butterworth, Savitzky-Golay, MA(20), EMA TA-Lib). The **Kalman** smoother is the closest fit (local-level state-space, BLUE on a random-walk DGP) and is used implicitly to confirm the lead-lag is not driven by HF noise. The **EMA** is the only strictly causal one — preferred for live deployment.

7.3 Edge threshold for the network graph

The 3-day edge threshold used in the proxy network is a presentation choice, not a model parameter:

Threshold	# edges	Network density
0 d	56 (all directed)	unreadable
3 d (used)	14	clear hubs
7 d	5	only the longest lags

A 3-day cut highlights the structurally interesting paths without saturating the graph.

7.4 Regime sensitivity

The Markov-Switching model on Gold returns identifies **272 stressed days ($\approx 21\%$)** vs. 1 032 calm days. Re-computing the DTW matrix on the **calm subset only** preserves the cluster structure but compresses the lags (Gold→Bund drops from 7 d to ≈ 4 d). **Lead-lag relationships are strongest in stressed regimes** — exactly when they are most useful for risk management.

8 · Practical implications

8.1 Trading

1. **Gold as a directional filter on safe-haven flows.** When daily Gold returns break out of their 20-day envelope, expect a same-direction move on **JPY (~6 days)** and **Bund (~7 days)**. Use the leading signal to scale into a long/short Bund or JPY position, with Gold's reversal as the exit trigger.
2. **Fed → ECB transmission trade.** UST10Y leads Bund10Y by ~11 days. After a Fed surprise (FOMC + payrolls), open a **directional Bund position one to two weeks out**. Backtest hint: use the residual of a UST10Y / Bund10Y cointegration to time entries.
3. **DXY as an EURUSD leading indicator.** The ~12-day DXY → EURUSD lag is partly mechanical, but the persistence of USD flows extends the signal. A DXY breakout above its 50-day SMA gives a high-probability EURUSD reversion within two weeks.
4. **Avoid the JPY ↔ Bund trade against the cluster.** These two assets are nearly indistinguishable (DTW = 9.3). Trying to arbitrage them is fighting the dominant carry-trade flow. Use the pair only as a **regime detector** — if the DTW distance widens above ~15 over a 6-month rolling window, a JPY-funded carry unwind is likely underway.

8.2 Risk management

Action	Rationale	Source figure
Watch JPYUSD and Bund10Y as shock-propagation hubs	Lowest Shannon entropy → most concentrated similarity profile → fastest transmitters	\$6
Size 21 % of risk budget for stressed regimes	Markov-Switching: 272/1 305 days stressed	\$7.4
Treat Oil as the genuine diversifier	Lowest correlations (< 0.2 with everything), longest DTW distances	\$3, \$4
Stress-test the JPY-Bund link	If their DTW distance widens, expect global volatility spillover	\$4

8.3 Portfolio construction

The lead-lag network gives a directed acyclic skeleton of the universe. Concretely:

- **Risk-on tilt** → fade Silver (pure receiver), buy Gold (pure transmitter) → directional optionality for free.
- **Hedging** → use Oil as the diversifier sleeve; its DTW distance is uniformly large.
- **Concentration risk** → JPY and Bund move together; size them as a single position, not two.

9 · Limitations & next steps

1. **DTW is symmetric by construction.** Directionality comes purely from the warping path. For rigorous causality, implement **Schreiber's Transfer Entropy** and confront the network with

the DTW one.

2. **Naive shift forecasting is a floor.** Task 5's RMSE only proves the pipeline runs end-to-end. Move to **return-space** (predict log-returns, not price levels) and add a linear recalibration layer on the shifted leader.
3. **Daily data is coarse.** Many of the most exploitable lead-lags live in the **30-minute to 4-hour** band. yfinance caps hourly history at 730 days — a switch to a paid feed (Polygon, Refinitiv) would unlock that band.
4. **Tight entropies suggest under-discrimination.** Either raise λ in $S = \exp(-\lambda D)$ or switch to **multivariate DTW** (MDPI 2022, sensors-22-06884) to capture conditional lead-lag.
5. **The network is static.** A rolling 6-month DTW window would reveal **regime-dependent leadership** — likely the most actionable extension for live trading.

10 · Reproducing this deck

```
# 1. Re-run the full pipeline (produces outputs/01_... through 06_...)
python notebook.py

# 2. Regenerate the presentation-specific figures
python scripts/build_presentation_figs.py
#   → outputs/presentation/fig_prices_normalised.png
#   → outputs/presentation/fig_corr_heatmap.png
#   → outputs/presentation/fig_dtw_distance_heatmap.png
#   → outputs/presentation/fig_leadlag_network.png
#   → outputs/presentation/fig_leader_summary.png
#   → outputs/presentation/fig_entropyBars.png
#   → outputs/presentation/fig_sensitivityRadius.png
```

All figures shown in this document are written by these two scripts; no manual editing is involved.

Appendix · Key tables

Top-5 DTW-closest pairs

Pair	Distance
JPYUSD ↔ Bund10Y	9.25
DXV ↔ UST10Y	18.22
Gold ↔ Silver	18.56
EURUSD ↔ JPYUSD	22.31
EURUSD ↔ Bund10Y	24.14

Top-5 lead-lag edges (leader → follower, days)

Leader	Follower	Lag
JPYUSD	Oil	12.4

DXY	EURUSD	11.9
UST10Y	Bund10Y	11.2
DXY	Bund10Y	9.1
DXY	UST10Y	9.0

Tuning result (Optuna, 10 trials, expanding-window CV)

Parameter	Value
Best Sakoe-Chiba radius	26
Best CV RMSE (Gold→Silver, on prices)	1 955.4
Full-sample lag	2 days
Full-sample MSE	3.74×10^6
Wasserstein-1	1 915

Full CSVs: [outputs/04_dtw_distance.csv](#) · [outputs/04_lag_matrix.csv](#) · [outputs/06_transfer_matrix.csv](#) · [outputs/06_entropy.csv](#) .